

Complex system: many interacting components whose collective behavior is qualitatively different from and not predictable from the parts in isolation. A complicated system (e.g., a jet engine) can be understood by disassembly; a complex system (e.g., an economy) cannot – the interesting properties live in the interactions.

Emergence: global properties absent from and unpredictable from individual components (e.g., flocking from Boids’ three rules; gliders from Game of Life’s four rules; segregation from Schelling’s mild individual preferences).

Boids: (1) *Separation* – avoid crowding neighbors; (2) *Alignment* – match average heading; (3) *Cohesion* – move toward average position. Global flocking from purely local rules; no central controller.

Three modeling approaches: *Mathematical (equations)* – analytically tractable, well-mixed, misses spatial structure. *ABM* – captures heterogeneity, spatial effects, emergence; parameter-heavy. *Network* – captures interaction topology; costly to define. Trade-offs generalise: equations give global insight at the cost of abstraction; ABM/network add realism at the cost of complexity.

Model taxonomy (classify on all four axes): *time* = discrete vs. continuous; *linearity* = linear (no cross-terms) vs. nonlinear; *autonomy* = autonomous (no explicit t) vs. non-autonomous; *order* = number of state variables. Bank account: discrete, linear, autonomous, 1st. Logistic map: discrete, nonlinear, autonomous, 1st. Lotka-Volterra: continuous, nonlinear, autonomous, 2nd. Lorenz: continuous, nonlinear, autonomous, 3rd.

Reduction trick: any non-autonomous system becomes autonomous by adding $\dot{\tau} = 1$ ($\tau = t$ as extra variable). Any n th-order ODE becomes 1st-order by introducing $y_i = x^{(i-1)}$. Both conversions are exact.

Fixed point (discrete): x^* s.t. $f(x^*) = x^*$; geometrically the intersection of $y = f(x)$ with the diagonal $y = x$. **Stability:** stable iff $|f'(x^*)| < 1$; unstable iff $|f'(x^*)| > 1$; marginal at $|f'| = 1$.

Cobweb diagram: start at (x_0, x_0) ; go vertical to $(x_0, f(x_0))$; go horizontal to $(f(x_0), f(x_0))$; repeat. Spiral inward: stable. Spiral outward: unstable. Staircase: monotone convergence.

Basin of attraction: set of initial conditions from which the orbit converges to a given stable FP. Unstable FPs have no basin; they repel all nearby trajectories.

Logistic map: $x_{n+1} = rx_n(1 - x_n)$, $r \in [0, 4]$. $r < 1$: extinction. $1 < r < 3$: stable FP at $x^* = 1 - 1/r$. $r = 3$: first period-doubling bifurcation. Cascade with decreasing interval ratios. $r \approx 3.57$: onset of chaos. Chaotic band (with periodic windows) up to $r = 4$.

Bifurcation diagram: long-run attractor values plotted vs. parameter r . At each period-doubling: $\lambda = 0$. In stable windows: $\lambda < 0$. In chaotic band: $\lambda > 0$. The Lyapunov exponent $\lambda(r)$ decodes the diagram’s structure.

First-order linear discrete: $x_{t+1} = ax_t + b$ has closed-form solution $x_t = a^t x_0 + b(1 - a^t)/(1 - a)$ for $a \neq 1$, or $x_t = x_0 + bt$ for $a = 1$. Always analytically solvable.

Fixed point (continuous): x^* s.t. $f(x^*) = 0$. Stable (attractor) iff $f'(x^*) < 0$. Unstable (repeller) iff $f'(x^*) > 0$.

1D autonomous continuous cannot oscillate: in $\dot{x} = f(x)$ with $x \in \mathbb{R}$, the phase line is monotone – trajectories cannot reverse without crossing a FP. The statement “ $\dot{x} = f(x)$ cannot oscillate whatever f is” is *true for 1D* but *false for $n \geq 2$* (Lotka-Volterra oscillates). Non-autonomy also breaks it. This nuance is the most common exam trap.

Euler: $x_{t+\Delta t} = x_t + \Delta t f(x_t)$; error $O(\Delta t)$ per step; fast, inaccurate.

RK4: 4 evaluations of f per step; error $O((\Delta t)^4)$; far more accurate for smooth f .

Lotka-Volterra: $\dot{x} = ax - bxy$, $\dot{y} = -cy + dxy$. FPs: $(0, 0)$ unstable saddle; $(c/d, a/b)$ center surrounded by closed periodic orbits. Prey and predator oscillate out of phase; amplitude set by initial conditions.

Romeo-Juliet: $\dot{R} = aR + bJ$, $\dot{J} = cR + dJ$. Signs of a, b, c, d encode each person’s response to own vs. partner’s feelings (e.g., $a < 0, b > 0$: retreats from own love, encouraged by partner’s). Long-term behavior from eigenvalues of the 2×2 system matrix.

Lorenz: $\dot{x} = \sigma(y - x)$, $\dot{y} = x(\rho - z) - y$, $\dot{z} = xy - bz$; canonical $\sigma = 10, b = 8/3, \rho = 28$. Originally a simplified convection model. First clear demonstration that a simple 3D deterministic system can produce aperiodic, practically unpredictable behavior.

$\dot{x} = x$: no chaos possible – 1D continuous; Poincaré-Bendixson requires $n \geq 3$ for continuous autonomous chaos; and this system’s solutions $x(t) = x_0 e^t$ are unbounded, violating the boundedness condition.

Four chaos conditions (all required; removing any one loses chaos): (1) *Deterministic* – no randomness; same IC yields same trajectory. (2) *Aperiodic* – orbit never exactly repeats; periodicity would restore predictability. (3) *Bounded* – orbit stays finite; unbounded divergence is trivial, not chaos. (4) *SDIC* – nearby initial conditions diverge exponentially; this is the operational signature of chaos.

Lyapunov exponent: $\lambda = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=0}^{n-1} \ln |f'(x_i)|$. $\lambda > 0$: chaos (SDIC). $\lambda < 0$: stable orbit. $\lambda = 0$: at a bifurcation point or onset of chaos.

Ergodicity: time averages equal ensemble averages on the attractor; long-run statistical properties are stable even when individual trajectories diverge. Permits statistical prediction of chaotic systems (climate, not weather).

Strange attractor: bounded, aperiodic, fractal structure (non-integer Hausdorff dimension), SDIC along it. Example: Lorenz butterfly, $D \approx 2.06$.

Feigenbaum constant: $\delta \approx 4.669 = \lim_n (r_n - r_{n-1}) / (r_{n+1} - r_n)$. Universal: the same value for any smooth unimodal 1D map regardless of its specific form.

Chaos in continuous systems requires $n \geq 3$: by Poincaré-Bendixson, bounded trajectories in $\mathbb{R}^2$ converge to FPs or limit cycles. Chaos needs at least 3 dimensions; hence Lorenz.

Fractal: geometric object with self-similar dimension D strictly greater than its topological dimension d_T . d_T is always a non-negative integer (0=point, 1=curve, 2=surface); D is typically non-integer.

Self-similar dimension: $D = \log N_C / \log M_F$ (N_C = number of self-copies, M_F = magnification factor). Cantor set: $\log 2 / \log 3 \approx 0.631$. Sierpiński triangle: $\log 3 / \log 2 \approx 1.585$. Koch curve: $\log 4 / \log 3 \approx 1.262$.

IFS: a complete metric space + a finite set of contractive mappings $\{f_1, \dots, f_n\}$. Banach Fixed-Point Theorem guarantees a unique attractor A^* satisfying $A^* = \bigcup_i f_i(A^*)$; every starting set converges to it.

DIFS (deterministic): $S_{n+1} = \bigcup_i f_i(S_n)$ at each step. Applies all transformations to the entire current set; converges to the attractor.

RIFS / Chaos Game (random): at each step, pick one transformation at random (with assigned probabilities) and apply to the current *point*; plot the result. After a transient, plots trace the same attractor as DIFS, distributed by the invariant measure. Works because Banach’s theorem guarantees convergence to the unique fixed-point set regardless of how you sample it.

Sierpiński triangle – 3 methods: (1) geometric deletion (remove central upward triangle from each filled triangle repeatedly); (2)

DIFS (apply 3 scaling-by- $\frac{1}{2}$ contractions to the whole set simultaneously); (3) RIFS/Chaos Game (pick a random vertex of the equilateral triangle, move halfway to it, plot, repeat).

Applications: image compression (store IFS parameters, not pixels; exploits self-similarity). Fractal antennas (Koch curve boundary: high perimeter per area; self-similarity at multiple scales gives multiband resonance).

Julia & Mandelbrot (*absent from this year's slides; appeared in past exams*): iterate $f_c(z) = z^2 + c$ in $\mathbb{C}$.

Filled Julia set K_c : $\{z_0 \in \mathbb{C} : |f_c^n(z_0)| \not\rightarrow \infty\}$. Julia set $J_c = \partial K_c$ (boundary of K_c).

Critical orbit: orbit of $z_0 = 0$ (the critical point of f_c). If it escapes to ∞ : J_c is Cantor dust (totally disconnected). If it stays bounded: J_c is connected. This is the Julia-Fatou theorem.

Mandelbrot set $\mathcal{M}$: $\{c \in \mathbb{C} : |f_c^n(0)| \not\rightarrow \infty\}$, i.e., the c -values for which the critical orbit stays bounded. Catalog of Julia sets: $c \in \mathcal{M} \Leftrightarrow J_c$ connected; $c \notin \mathcal{M} \Leftrightarrow J_c$ Cantor dust.

Julia set via IFS: iterate inverse branches $\pm\sqrt{z-c}$ chosen randomly; orbit traces J_c (RIFS applied to the complex map).

Graph metrics: eccentricity $\varepsilon(u) = \max_v d(u, v)$; diameter $= \max_u \varepsilon(u)$; radius $= \min_u \varepsilon(u)$; center $= \{u : \varepsilon(u) = \text{rad}\}$; periphery $= \{u : \varepsilon(u) = \text{diam}\}$. Always: $\text{rad} \leq \text{diam} \leq 2\text{rad}$.

Clustering coefficient: $c_v = 2e_v/[k_v(k_v - 1)]$ where e_v = edges among v 's k_v neighbors. Example: 4 neighbors, 3 edges $\Rightarrow c_v = 6/12 = 0.5$.

Centrality: degree $= k_v/(n - 1)$ (hubs, popular nodes); closeness $= (n - 1)/\sum_u d(v, u)$ (fastest spreaders); betweenness $= \sum_{s \neq v \neq t} \sigma_{st}(v)/\sigma_{st}$ (bridges, bottlenecks).

Bipartite graph: vertices split into U, V with all edges between U and V . No odd-length cycles (bipartite $\Leftrightarrow$ no odd cycles).

Erdos-Renyi $G(N, p)$: each edge added independently with prob p ; Poisson degree distribution. Giant component ($O(N)$ nodes) appears at percolation threshold $\langle k \rangle \approx 1$ ($p_c \approx 1/N$): a phase transition in connectivity.

Barabasi-Albert: new nodes attach with m edges via preferential attachment (prob $\propto k$: "rich get richer"). Produces power-law degree distribution $P(k) \sim k^{-\gamma}$, $\gamma \approx 3$. Scale-free: a few hubs dominate.

Watts-Strogatz small-world: start from ring lattice (each node connects to k nearest neighbors); rewire each edge with prob p . At intermediate p : high clustering (retained from lattice) + short average paths (from random shortcuts). Two hallmarks: $C \gg C_{\text{rand}}$ and $L \approx L_{\text{rand}}$.

Scale-free vs. small-world robustness: scale-free is robust to *random* node removal (a random node is unlikely to be a hub) but fragile to *targeted attack* (removing hubs rapidly disconnects the graph). Small-world networks are more uniformly vulnerable, meaning targeted attacks are less catastrophic since no single hub is dominant. The statement "small-world is always more robust" is too absolute: for random failures, scale-free can be *more* robust.

Modularity Q : $\sum_c [L_c/m - (d_c/2m)^2]$; $Q \approx 0$: structure no better than random; $Q > 0.3$: strong communities. Girvan-Newman: iteratively remove the max-betweenness edge (recompute after each removal); dendrogram reveals hierarchy; stop when Q is maximised. Louvain: greedy Q -maximisation; faster, scales to large graphs.

Assortativity: tendency to connect to similar nodes. Degree assortativity $r > 0$: hubs connect to hubs (social networks). $r < 0$: hubs connect to low-degree periphery (internet, biological networks).

Voter model: each node copies the opinion of one random neighbor per step. Absorbing state: consensus. Slower on clustered networks; faster on random/small-world.

SIS (network): node i recovers at μ ; susceptible neighbor j gets infected at β . Epidemic threshold: $\beta/\mu > 1/\lambda_{\text{max}}(A)$. Scale-free nets have large λ_{max} , so a lower threshold – epidemics spread more easily than in homogeneous populations.

SIR (ODE, well-mixed): $\dot{S} = -\beta SI/N$, $\dot{I} = \beta SI/N - \gamma I$, $\dot{R} = \gamma I$. Epidemic grows iff $R_0 = \beta/\gamma > 1$.

Kuramoto: $\dot{\theta}_i = \omega_i + (K/N) \sum_j \sin(\theta_j - \theta_i)$. Order parameter $r = |N^{-1} \sum_j e^{i\theta_j}|$: $r \approx 0$ incoherent; $r \approx 1$ fully synchronised. Above critical K_c : spontaneous synchronisation (fireflies, power grids).

Diffusion: $\dot{c}_i = \alpha \sum_{j \in N(i)} (c_j - c_i)$. Topology determines equilibration rate; isolated or peripheral nodes lag.

CA formal: (S, s_0, N, d, f) – finite state set, initial configuration, neighborhood template, dimension, local rule applied synchronously. Rules count: $q^{2^{r+1}}$. Binary ($q = 2$), $r = 1$: $2^8 = 256$; $r = 2$: $2^{32} \approx 4.3 \times 10^9$.

Moore vs. Von Neumann (2D): Moore = 8 neighbors (cardinal + diagonal); VN = 4 cardinal only. Neither universally better: Moore gives more isotropic spreading; VN is simpler and sufficient when diagonal interactions don't matter. The correct choice depends on the physical reality being modeled – neither is "preferable" in the abstract.

Wolfram classes: (1) all cells reach a uniform state; (2) stable or periodic patterns; (3) chaotic aperiodic patterns filling space; (4) complex localised structures, long transients, computationally universal (e.g., Rule 110). Class 4 sits at the boundary of order and chaos.

Game of Life (2D binary, Moore): live cell survives with 2–3 live neighbors; dies otherwise. Dead cell with exactly 3 live neighbors is born. Static: block. Periodic: blinker. Mobile: glider.

CA rule tracing (*most repeated exam question*): for each cell i , extract the neighborhood triple (x_{i-1}, x_i, x_{i+1}) at time t and the observed $x_i(t+1)$. Check every triple: does the same triple ever map to two different outputs? If yes: the observed transition is *inconsistent* with any single deterministic rule – it is impossible. If no: reconstruct the rule table from all observed triples.

Schelling segregation: agents move when fewer than $t\%$ of neighbors share their type. Even $t \approx 30\%$ produces strong global segregation from mild individual preferences. Canonical emergence example: the collective outcome far exceeds what individual preferences would suggest.

ABM vs. ODE: ABM captures individual heterogeneity, stochasticity, spatial structure, local effects (e.g., epidemic clustering, stochastic extinction in small populations). ODE assumes homogeneous mixing, deterministic, mean-field. ABM reveals threshold behavior and pattern formation that ODEs miss.

ABM vs. network (predator-prey): ABM is more appropriate when spatial movement and local encounters matter (patch extinction, territory dynamics). Network approach is more appropriate when the key feature is *who interacts with whom* (a fixed food web topology, specific predation links) rather than spatial position.

SIR-ABM: agents $\in \{S, I, R\}$; $S \rightarrow I$ with prob β per infected contact; $I \rightarrow R$ with prob γ per step. Advantages over ODE-SIR: spatial spread, stochastic extinction, individual heterogeneity, multiple co-morbidities. Air-travel extension: city nodes + flight network; agents carry their state when traveling, enabling long-range epidemic seeding before local spread is detected.

Majority rule vs. voter model: both have consensus as an absorbing state. Voter: stochastic, asynchronous (copy one random neighbor); always reaches consensus on finite connected graphs. Majority rule: deterministic, synchronous (adopt majority opinion);

may freeze in non-consensus configurations on some topologies.

GNN message passing (per layer): (1) each node sends message $M(h_u)$ to each neighbor; (2) aggregate: $a_v = \text{AGG}(\{m_{uv} : u \in N(v)\})$; (3) update: $h'_v = U(h_v, a_v)$. After k layers: h_v encodes the k -hop neighborhood of v .

GCN: $H^{(k+1)} = \sigma(\tilde{A}H^{(k)}W^{(k)})$ (normalised adjacency). Uses the full graph at training; *transductive* – cannot generalise to unseen nodes without retraining.

GraphSAGE: samples a fixed-size neighborhood; learns an aggregation function (mean/LSTM/pool). *Inductive* – generalises to new

nodes or entirely new graphs without retraining. Scales to large and dynamic graphs.

Transductive vs. inductive: transductive = trained and tested on the same fixed graph (GCN). Inductive = learns a function applicable to unseen graphs (GraphSAGE). Transductive task: classifying nodes in a known citation network. Inductive task: predicting properties of new molecules not seen at training time.

GNN tasks: node classification (paper topic in citation graph); link prediction (friend recommendation, protein interactions); graph classification (molecule toxicity, brain connectivity).